

Sewon Park박세원

AI-native Platform & Product Engineering Lead

CAREER PROFILE

DEVICE BOUNDARY TO AI PLATFORM

Across roughly eight years, I have moved from telecom network software and radar-camera-FPGA systems to biometric access-control platforms, then to AI-native developer products and engineering transformation. My strength is connecting device packets and SDKs to backend services, web experiences, agent workflows, and evidence-based release gates.

EXPERIENCE	Approx. 8 years · 3 companies
CURRENT	Integration Development Team Lead
FOCUS	AI Platform · DevEx · Industrial Software
PRINCIPLE	Verification before trust

01 PROFESSIONAL HISTORY

EXPERIENCE

2021.02–PRESENT

Suprema

Seongnam, Korea

5+ YEARS

Platform & Product Engineering · Integration Development Team Lead

Lead cross-functional AI-native missions while retaining hands-on ownership of the access-control product core. Work spans C++ device packets and SDKs, Java/Kotlin services, identity and event domains, React products, cloud sandboxes, and verification systems.

2026.04–PRESENT **PRODUCTIZING**

BioStar Developer Portal · 1 person · 8 weeks

PROBLEM	API, SDK, documentation, and execution environments were fragmented, so a developer’s first success depended on setup knowledge and technical support.
ACTION	Owned competitor analysis, problem framing, architecture, product design, full-stack implementation, and verification. Unified Docs, API SSOT, a device SDK emulator, MCP tools, a Linux/AWS sandbox, and evaluation gates.
OUTCOME	Postman-free browser API execution, hardware-free SDK workflow rehearsal, and natural-language replay of user-door-authentication-event histories. Selected among five finalist teams, later received the company-wide excellence award, and is being expanded toward physical-device workflows for a September release target.

TypeScript · React · Python · REST API/SDK · MCP · Docker/Linux · AWS · Playwright

2026.07–PRESENT **IN PROGRESS**

~200K LOC modernization · frontend team lead

PROBLEM	Angular embedded in an iframe and mixed with React had accumulated routing, back-navigation, and state-boundary debt.
ACTION	Lead the frontend team through an AI-native rewrite and separate backend payload/response, API contract, unit, React/useEffect, DOM, UI, CSS assertion, and E2E checks into independent meaning-preservation gates.
OUTCOME	Removed the architectural causes of the navigation and state problems and established a repeatable method for validating AI-generated migration code toward the next major release.

TypeScript · React · Angular · Playwright · REST API · DOM/UI/CSS/E2E

2026–PRESENT **IN PROGRESS**

Version-aware E2E · release confidence as a shared asset

GOAL	Detect product- and version-specific regressions before release instead of finding them again through manual checks.
ACTION	Turn critical journeys into reproducible fixtures and layer API, UI, and CSS gates so the same scenario can be replayed against each release line.
OUTCOME	Building a common regression system that makes version differences visible and release confidence reusable across product teams.

TypeScript · Playwright · REST API · UI/CSS assertion · E2E automation

2021–2024 CORE PLATFORM		
Gateway, device, and distributed-event platform		
SCOPE	Integrated device packets, capability cache, selective user sync, biometric login, face registration, Remote Enrollment, and new-hardware features across C++ servers, Java/Kotlin services, and web products.	
PLATFORM	Replaced Unified Gateway/Nginx with Spring Cloud Gateway, implementing URL and header routing, rate limiting, circuit breaking, TLS 1.2, and dynamic configuration refresh without restarts.	
OUTCOME	Improved distributed-event processing by 40%+, reduced real-time lookup CPU time by about 50%, and shortened bulk-user deletion from the 30-second range to about three seconds (30 sec → 3 sec).	
Java · Kotlin · C++ · Spring Boot · MariaDB · Microsoft SQL Server · gRPC · Apache Ignite · ZooKeeper · AWS Lambda/S3		
2025–2026 ENGINEERING ENABLEMENT		
AI-native delivery and verification practices		
ACTION	Designed a GitLab MR event → coverage measurement → LLM unit-test generation → coverage remeasurement workflow and led three interns through its implementation.	
SCALE	Shared agent architecture, AI-native workflow, and verification-gate practices through the TA AI TF, internal seminars, examples, and mentoring.	
OUTCOME	Turned individual AI-assisted implementation habits into workflows that teams and learners can repeat and verify.	
GitLab CI/CD · LLM agents · code coverage · unit-test automation · evaluation gates		
2020.04–2021.01 10 MONTHS	HBrain Research Engineer · Sensor Software	Korea
- Built radar-camera-FPGA pedestrian-detection and visualization software with C/C++, MFC, OpenGL, Windows API, and device SDKs; debugged sensor communication and integrated new device functions. - Authored requirements, protocols, and operational scenarios for public R&D projects and presented AI traffic-safety work at the Korean Society of Intelligent Transport Systems.		
2018.10–2020.02 1 YEAR 5 MONTHS	AtData Software Developer · Network Management	Korea
- Developed and operated SK Telecom TANGO network-management web software, including IP configuration, transmission-network visualization, and raw-data processing. - Supported change requests, built new server environments, and automated delivery with Jenkins.		
2020	ACADEMIC FOUNDATION	
EDUCATION & RESEARCH		
2026–PRESENT IN PROGRESS	Ph.D. coursework · AI Convergence Sungkyunkwan University	
2019–2021 M.S.	Applied Data Science Sungkyunkwan University Thesis: response-target prediction for public petitions under class imbalance	
2010–2017 B.S.	Maritime Police Systems Gyeongsang National University	
ACTIVE RESEARCH	Explainable traditional-dance retrieval & recommendation Combines expert domain knowledge and VLM evidence to retrieve, recommend, and explain movement candidates without hiding the basis of judgment.	

ACTIVE RESEARCH

Hybrid quantum-classical model evaluation

Compares classical ML and QML candidates under shared splits, budgets, and evaluation gates so model choice is based on evidence rather than novelty.

03

LEADERSHIP & COMMUNITY

ENABLEMENT & RECOGNITION

2025-2026

MENTORING

Four AI learning programs · sustained project mentoring

Supported problem framing, technology choices, implementation, and verification across Codeit, an AI bootcamp, and SK Networks Family AI Camp cohorts 27 and 28. Current teams cover a traffic-dispute AI service and a personalized fashion-recommendation service.

2025-2026

INTERNAL LEADERSHIP

Three interns · AI verification workflow

Led implementation of LLM-assisted test generation and shared verification-first engineering through the TA AI TF, internal seminars, and reusable examples.

2026.06

EXCELLENCE AWARD

Oldboy Project · BioStar Developer Portal

Selected as one of five finalist teams with KRW 3 million in development funding; the completed AI-native product later received the company-wide excellence award with a KRW 10 million prize.

최종 5팀에 선정되어 개발비 300만원을 받은 뒤 AI-native 제품을 완성했고, 전사 우수상과 상금 1,000만원을 받았습니다.

2026.01

INNOVATION AWARD

AI category · Encouragement Prize

Recognized for applying AI throughout 2025 to improve organizational productivity and technical competitiveness.

04

TECHNICAL FOUNDATION

SELECTED STACK

Platform & product	Java · Kotlin · Python · JavaScript/TypeScript · Spring Boot · React · REST · gRPC
AI & evaluation	LLM · MCP · RAG · agent workflows · PyTorch · contract/unit/DOM/UI/CSS/E2E testing
Device & identity	C/C++ · device SDKs · packet debugging · biometric authentication · user/door/authentication/event flows
Data & infrastructure	MariaDB · Microsoft SQL Server · Apache Ignite · ZooKeeper · AWS Lambda/S3/EC2 · Docker/Linux · Jenkins · GitLab CI/CD